

Modeling Community & Species Level Acoustic Responses to the 2024 Eclipse

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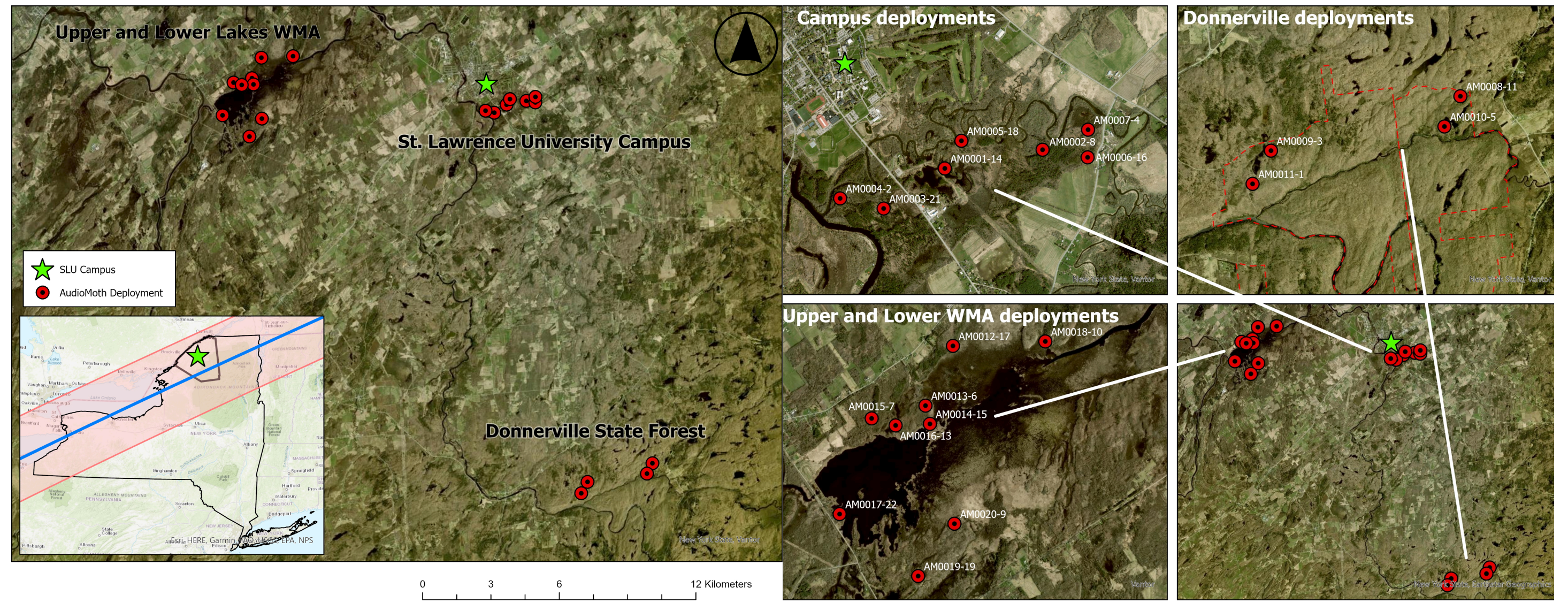
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Abstract

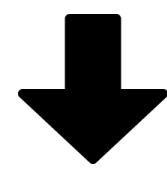
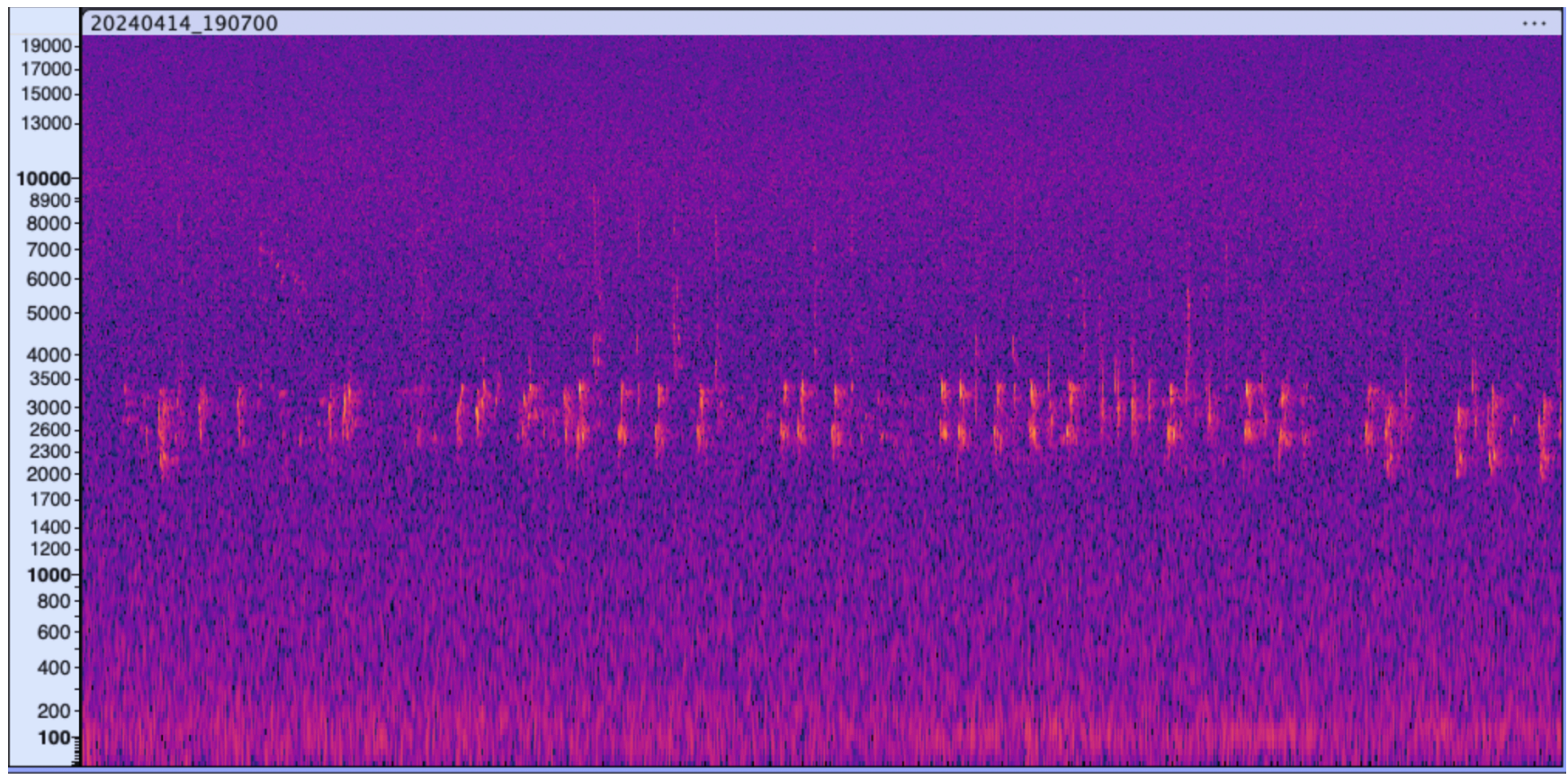
- Investigating changes in wildlife vocalization patterns due to solar eclipse on April 8th, with totality at 3:23 - 3:26 pm.
- Used an AI species identifier (BirdNET) to obtain species-level data.
- Goal: model community and species level response to eclipse using prescence/absence data.

Background

- Deployed 20 audio recorders (AudioMoths) in St. Lawrence County, NY.



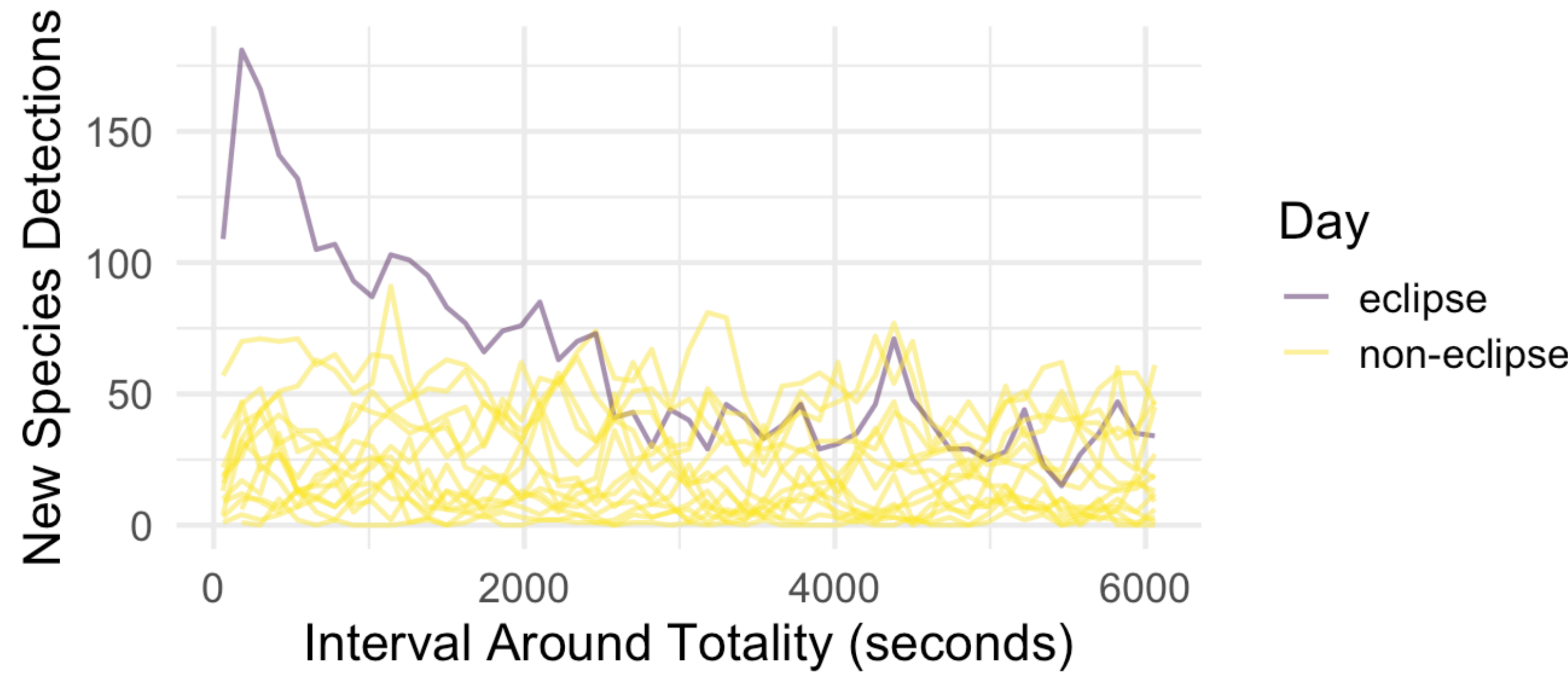
BirdNET Workflow



file	start	end	species	probability
20240401_232400.WAV	0	3	Pseudacris crucifer_Spring Peeper	0.8624
20240331_154000.WAV	30	33	Anas platyrhynchos_Mallard	0.7498
20240331_152500.WAV	36	39	Lithobates sylvaticus_Wood Frog	0.4021

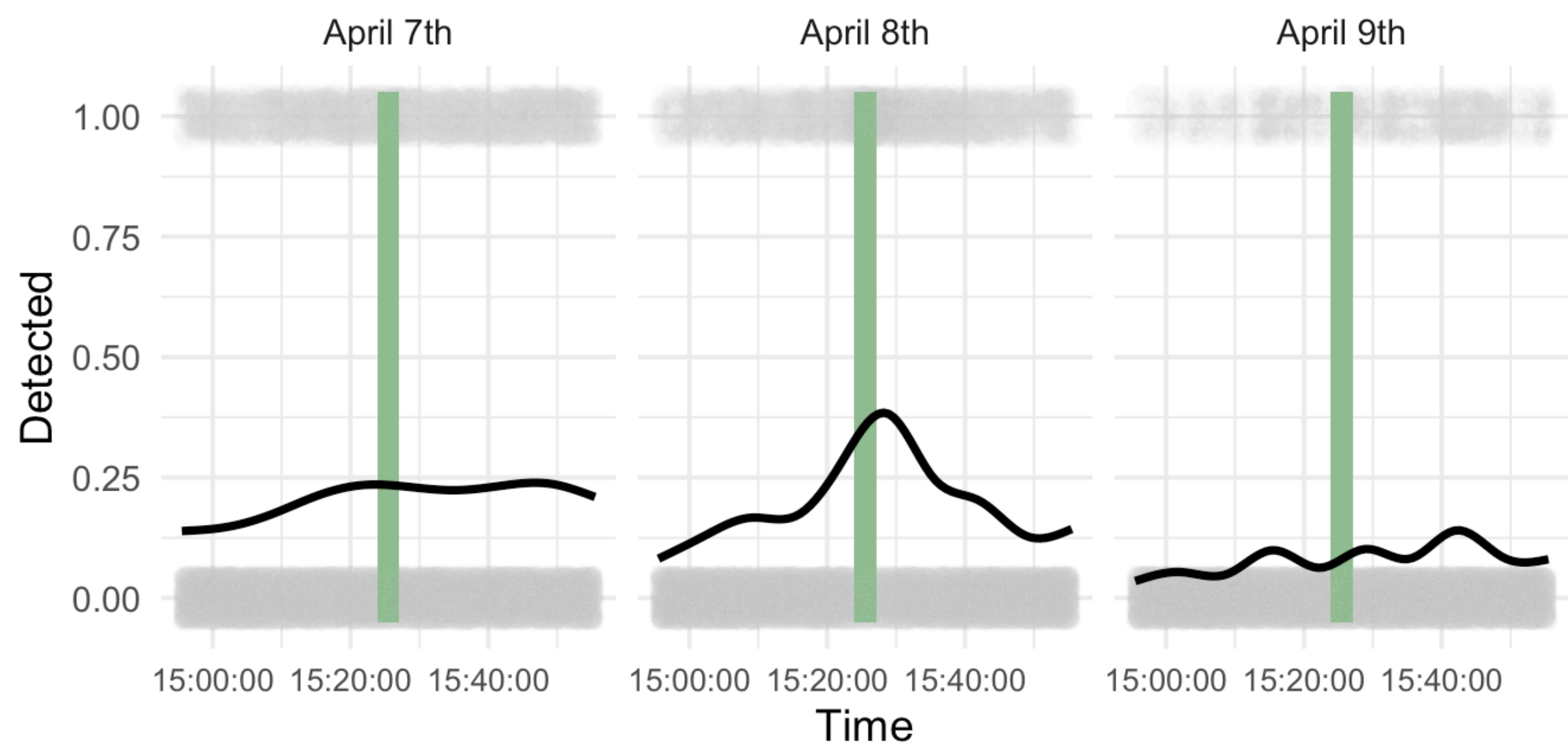
Data Exploration

Determining time interval of interest:



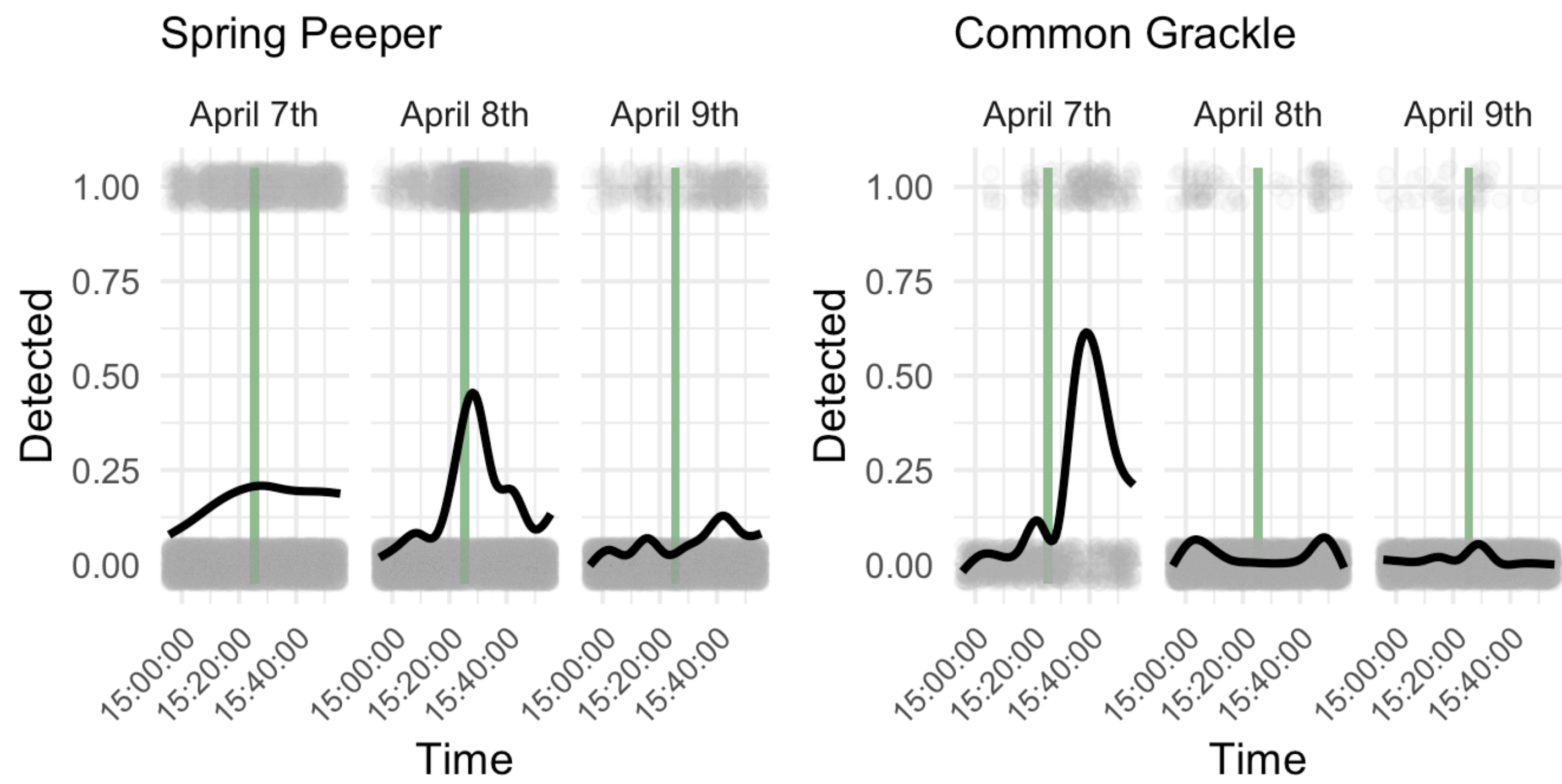
Community level graph:

- Community-level vocal activity increased during eclipse, peaking at totality.



Species level graphs:

- Evidence of species behaving differently due to the eclipse.

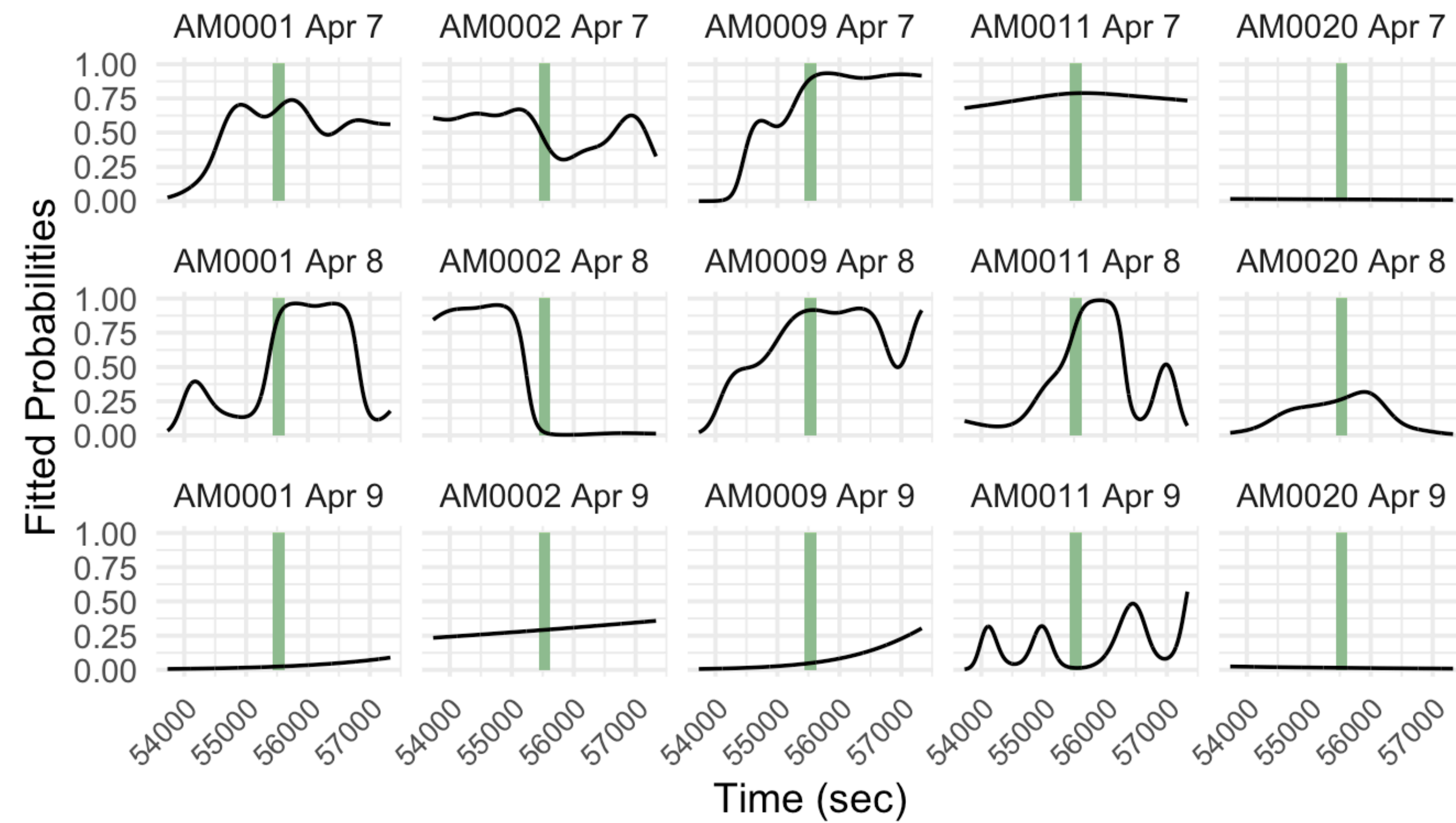


Modeling

Generalized Additive Mixed Model (GAMM):

$$\text{logit}(\pi_{iad}) = \beta_{ad} + f_{ad}(t_{iad}) + \epsilon_{iad}$$

- π_{id} is the probability of detecting a species for the i^{th} observation recorded on the a^{th} audiomoth ($a = 1, 2, \dots, 6$) on the d^{th} day of recording ($d = 1, 2, 3$).
- β_{ad} is the intercept for each audiomoth-day combination.
- $f_{ad}(t_{iad})$ is a smoother term for time (indexed by i) that changes for each audiomoth-day combination.
- t_{iad} is the time (in seconds) that the i^{th} observation was recorded at the a^{th} audiomoth and the d^{th} day.
- ϵ_{iad} are random errors that follow an AR(1) temporal correlation structure for each audiomoth-day combination with common shared correlation parameter p .



Takeaways

- Generally, increased detections during eclipse.
- Estimated detection is species-specific and location-specific.

Future directions:

- Run model for all sites and all days.
- Trends for similar species (i.e. anurans, birds).
- Validation of BirdNET output.

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BirdNET was developed by Cornell Lab of Ornithology.